RISHIKA PATEL

+91 7985925729 | rishika.patel2419@gmail.com | linkedin.com/in/rishika-patel | github.com/Rishikaaz

SUMMARY

Performance-driven CS undergraduate specializing in Scalable Backend Development and System Design. Expert in DSA, Python/C++ microservices, and cloud infrastructure (AWS). Demonstrated success in developing secure, containerized applications and resolving intricate algorithmic challenges.

EDUCATION

Vellore Institute of Technology (VIT) | B.Tech, CS (CGPA: 8.84) **2023 – 2027**
St. Xavier's Convent School | Senior Secondary (XII), ICSE 2022
St. Xavier's Senior Secondary School | High School (X), CBSE (91.4%) 2020

TECHNICAL SKILLS

Languages: C++, Python, C, C#, SQL, Bash Scripting
Backend & System Design: Microservices Architecture, Distributed Systems, Multithreading, Scalable System Design, Object-Oriented Programming, React.js, Typescript.
Tools & Frameworks: Docker, Docker Compose, Git, GitHub, Linux Shell Scripting, ELK Stack
Cloud & Infrastructure: AWS (EC2, S3, IAM, VPC), CI/CD Fundamentals, Containerization, Server Management
Core Concepts & Networking: Data Structures & Algorithms, TCP/IP, Socket Programming, Network Protocols, Secure Coding, System Optimization

PROJECTS

Containerized HIDS & Distributed SIEM Pipeline

Python, Docker, ELK Stack, Scapy, Linux | May 2026

- Spearheaded a containerized SIEM architecture using Docker Compose, optimizing host CPU/RAM overhead by reducing resource consumption by 40% through strategic VM migration.
- Developed a high-throughput Python HIDS sensor using Scapy to capture live traffic and stream real-time JSON alerts over TCP.

Carbon-Emission-Prediction & Data Pipeline

Python, Scikit-Learn, Pandas, NumPy | Jun 2025 – Jul 2025

- Built a predictive pipeline using Python and Scikit-learn to forecast environmental metrics from industrial datasets.
- Implemented feature engineering and data normalization pipelines to achieve highly accurate algorithmic predictions.
- Designed visualization modules to convert multi-dimensional software logs into actionable analytical insights.

AI-Powered Behavioral Anomaly Detection

Python, Scikit-learn, Pandas, Scapy, Networking | ML & Cybersecurity Project | Jul -2026

- Built an end-to-end threat detection pipeline that captures raw network packets using Scapy, converts them into behavioral features, and passes them to unsupervised models.
- Trained the system on baseline network traffic to identify subtle zero-day anomalies and multi-stage attack patterns that traditional signature-based rules miss.
- Streamed real-time JSON alert logs directly into SIEM workflows, cutting down false positives and enabling immediate incident response by 60%.

EXPERIENCE

Green Skills with AI Technologies Intern | Edunet Foundation ***Jun 2025 – Jul 2025 | Remote***
Engineered data preprocessing and feature selection pipelines in Python for environmental datasets, reducing model training variance and improving report generation efficiency by 25%.

Cyber Security and Ethical Hacking Intern | Tamizhan Skills ***Jun 2025 – Jul 2025 | Remote***
Programmed 6+ proof-of-concept Python security tools, including a custom multi-port socket scanner, file-encryption utilities, and AES-encrypted client-server communication channels.

ACHIEVEMENTS

- Solved 300+ LeetCode problems and actively competed in Codeforces contests with a max of 902 contest rating.
- Selected for the DSCI Cyber For Her Hackathon, qualified for Round 2 of Scripted By{Her} 2.0, Meesho Hackathon as well as in Honeywell and the APCSIP-2026 for a Cyber Cell internship..
- Ranked in the top 2% globally on TryHackMe with a 177-day learning streak in penetration testing and digital forensics.
- Placed 355th out of 5,947 global competitors in the TryHackMe Industrial-Intrusion CTF.
- Secured 7th rank nationally in the Shell n'Zen CTF, solving complex web security and reverse engineering challenges.

CERTIFICATIONS

- Blockchain & Applications (NPTEL, IITK)
- Cyber Threat Management (CISCO)
- Certified Cybersecurity Educator Professional (CCEP)
- Computer Networking (Coursera)
- AWS Cloud Practitioner